Differential Calc

D-07.07.25

* Sets are a collection of well-defined objects.

Two sets may have relation among them.

A function is a unique kind of relation where each input must have a unique output.

(a) $\{(1,4),(2,5),(3,6),(4,7)\}$

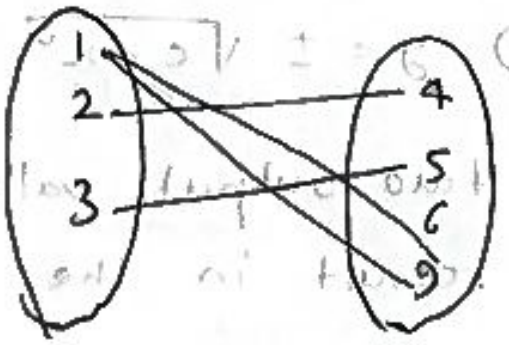
This is a function

(b) $\{(1,4),(2,5),(3,6),(1,4)\}$

This is a function

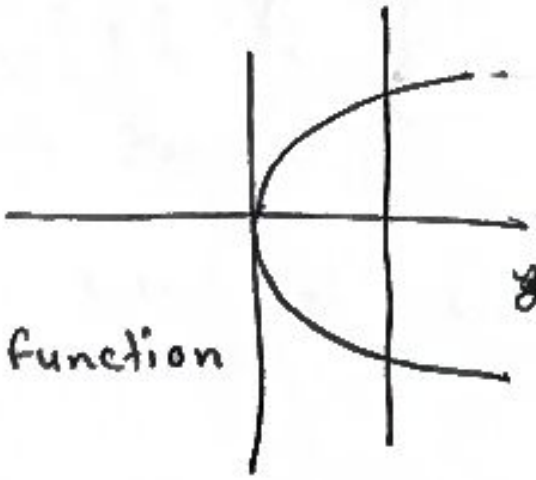
(c) $\{(1,4),(2,5),(3,6),(1,9)\}$

This is not a function ✗



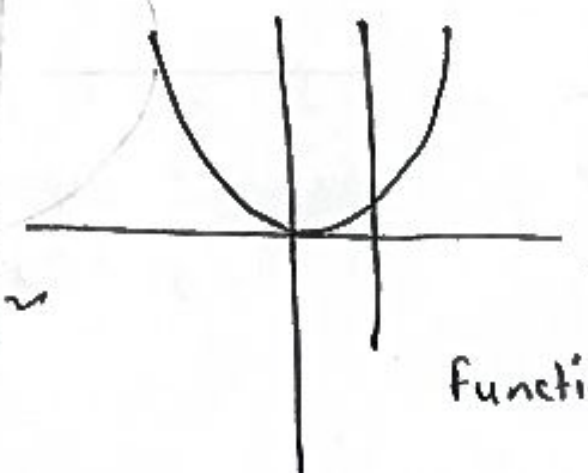
This violates the condition

To determine if an equation is a function, we may conduct vertical line test.



not a function

$x = y^2$



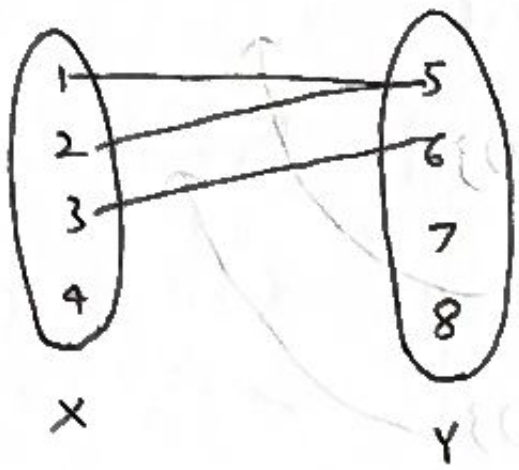
$y = x^2$

function

It's easy to determine a function from easily using a 'child-mother' technique

(my favorite since the $-\infty$ age)

<u>Example-01</u>

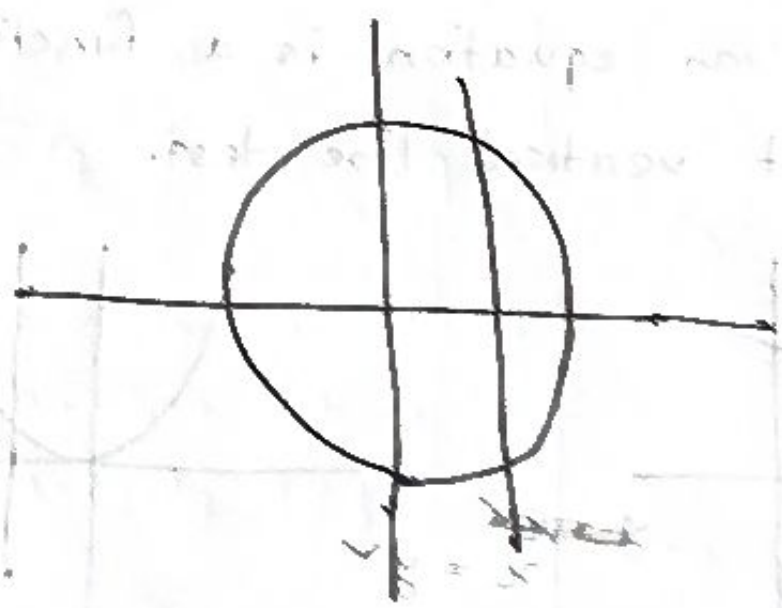


It's not a function; because the input 4 doesn't have 'validity'

* Function / equation of a circle isn't a function;

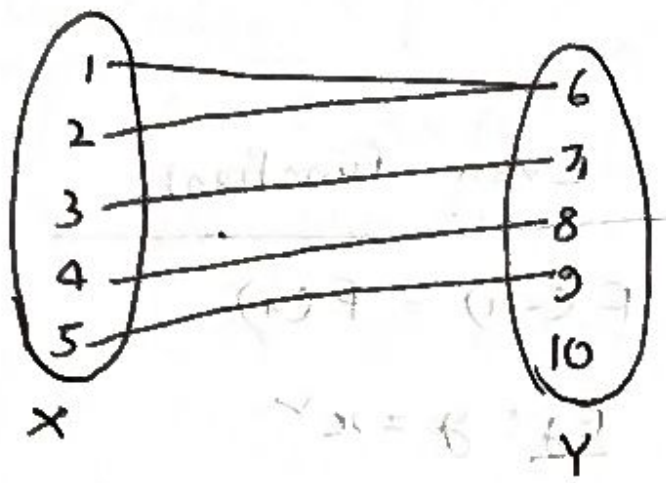
$x^2+y^2=c \Rightarrow y = \pm\sqrt{c-x^2}$

It will generate two output values for an input which will also result in the violation of the vertical line test.



$X \longrightarrow Y$

#

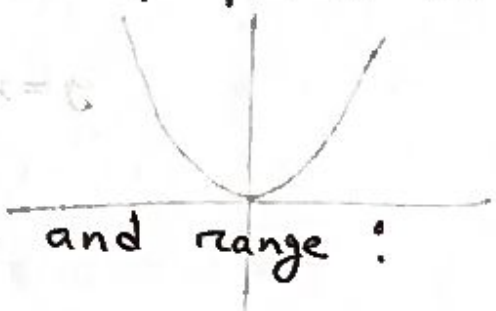


X is domain set

Co-domain set is Y

but the range set here is different

* Reading the properties of a graph is extremely important.



Domain and range :

(a) $y = 2x - 5$

$D = \mathbb{R}, \; R = \mathbb{R}$

(b) $F(x) = \sqrt{4 - 3t}$

$4 - 3t \geq 0$

$\Rightarrow t \leq \frac{4}{3}$

$\therefore$ Domain $= (-\infty, 4/3]$

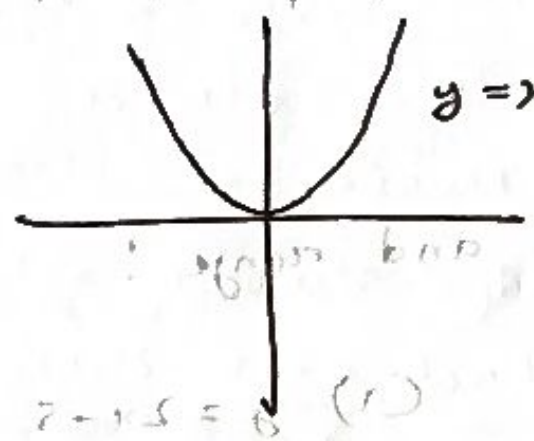
(c) $F(x) = \dfrac{\sqrt{3x+12}}{x-5}$

$x - 5 \neq 0$ & $3x + 12 \geq 0$

$\Rightarrow x \neq 5$ & $x \geq -4$

$\therefore$ Domain $= [-4, 5) \cup (5, \infty)$

Odd and even functions:

Odd functions:	Even functions
• $F(-x) = -F(x)$	• $F(-x) = F(x)$
Eg: $y = 5x^3 - x$	Eg: $y = x^2$
$F(-x) = -5x^3 + x$	• Symmetrical across the y axis
$= -F(x)$	
• Symmetric across the origin (y = x line)	

Difference quotient:

$$F(x-h) - F(x) / h \; ; \; h \neq 0$$

• This thing seems similar to the limit part

Rational and irrational numbers:

Rational numbers	Irrational numbers
p/q → form; $q \neq 0$	Eg: $\sqrt{3}, \pi, e,$
# $3.\overline{3}$ is a rational number which is actually $1/3$	$\sqrt{11}, 2+\sqrt{3}$
	no p/q form

Golden ratio:

$$\frac{1+\sqrt{5}}{2}$$

it's an irrational number too.

Trig function domain and range:

	Domain	Range
$\sin x$	$\mathbb{R}$	$[-1,1]$
$\cos x$	$\mathbb{R}$	$[-1,1]$
$\tan x$	$x = \frac{(2n+1)\pi}{2}$ বাদে $\mathbb{R}$	$\mathbb{R}$
$\operatorname{cosec} x$	$x = n\pi$ বাদে $\mathbb{R}$	$(-\infty,-1] \cup [1,\infty)$
$\sec x$	$x = \frac{(2n+1)\pi}{2}$ বাদে $\mathbb{R}$	$(-\infty,-1] \cup [1,\infty)$
$\cot x$	$x = n\pi$ বাদে $\mathbb{R}$	$\mathbb{R}$

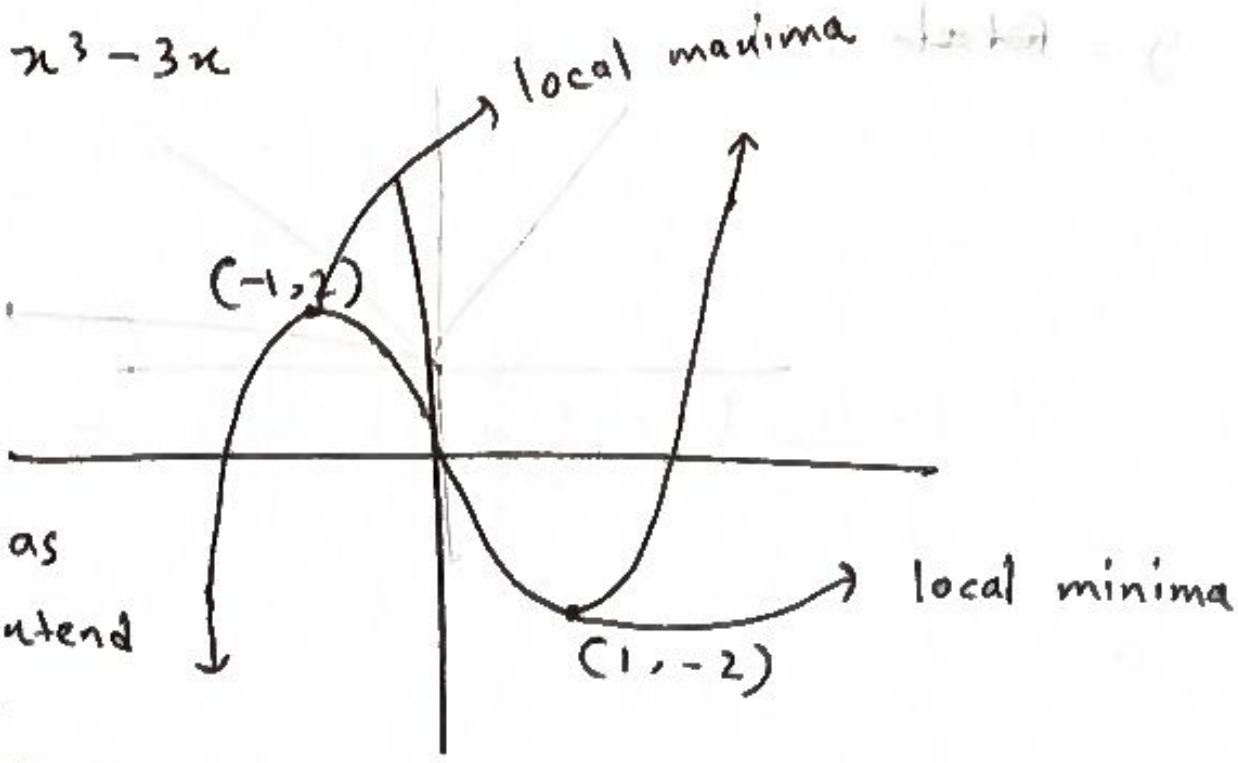
n is an integer

Local minima/maxima &

Global

- ~~Local~~ minima/maxima

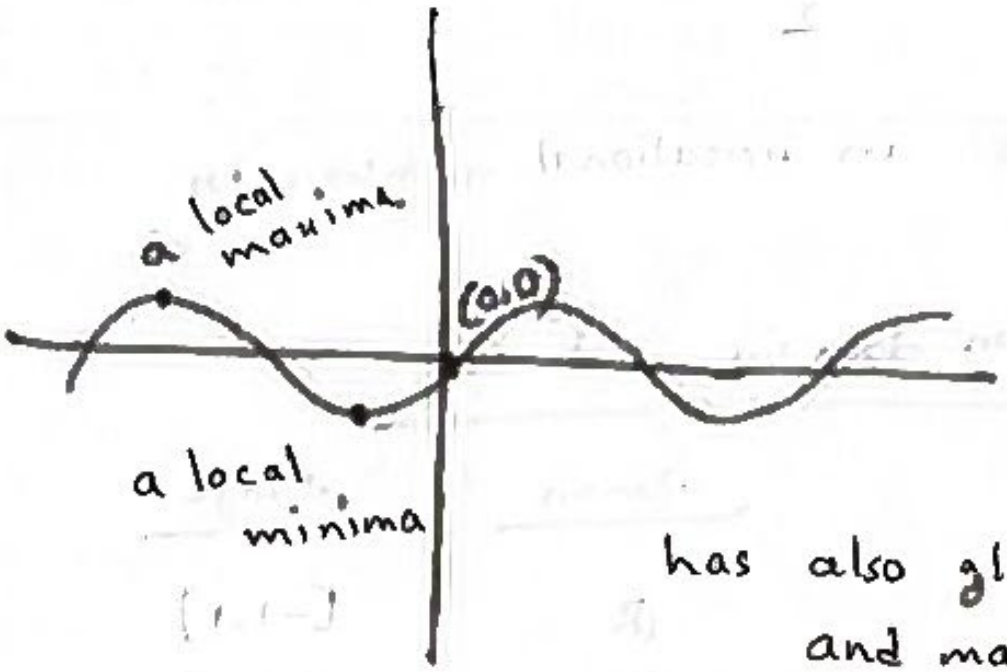
$F(x) = x^3 - 3x$



but NO

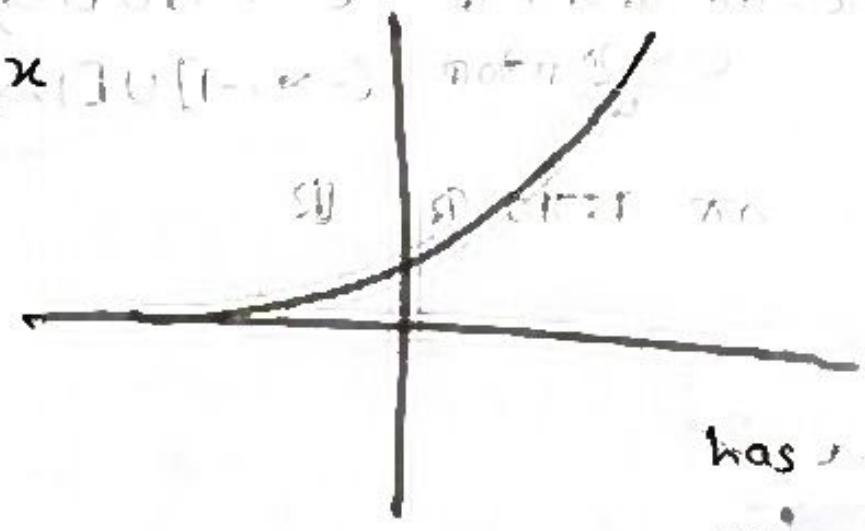
absolute highest or lowest point as they tend to extend to infinity

$F(x) = \sin x$



has also global minima and maxima as the absolute highest/lowest waves ever don't cross 1 or −1

$y = e^x$



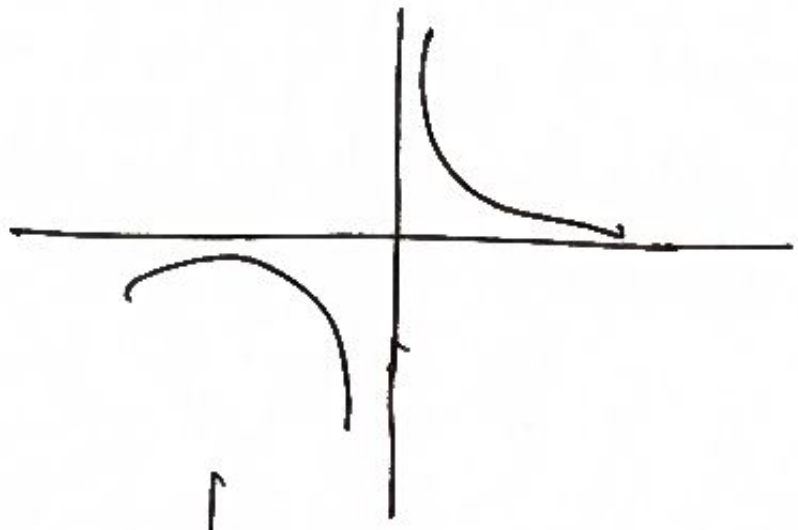
has no ~~local~~ kinds of minima and maximas

$y = |x|$

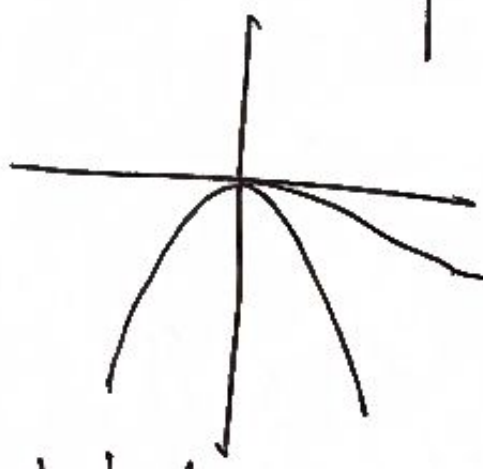
no maximas

local and global minima

$y = 1/x$ ⟶ no maxima or minimas



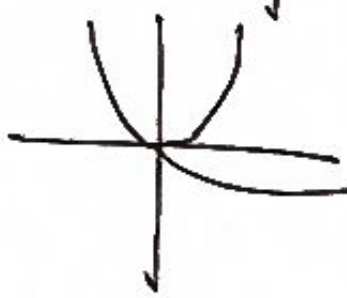
$y = -x^2$



no minima

both maximas

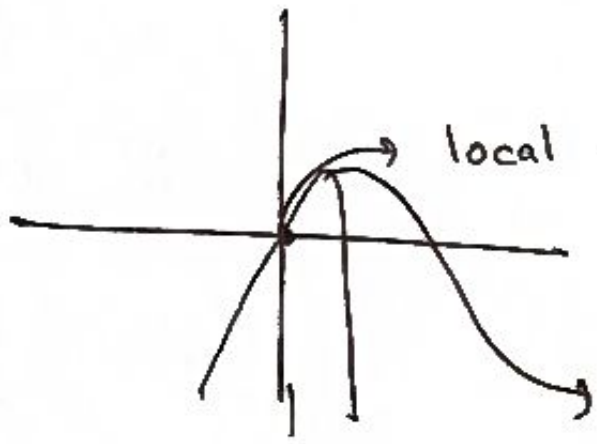
$y = x^2$



no maxima

both minimas

$y = -x^4 + 3x^2$

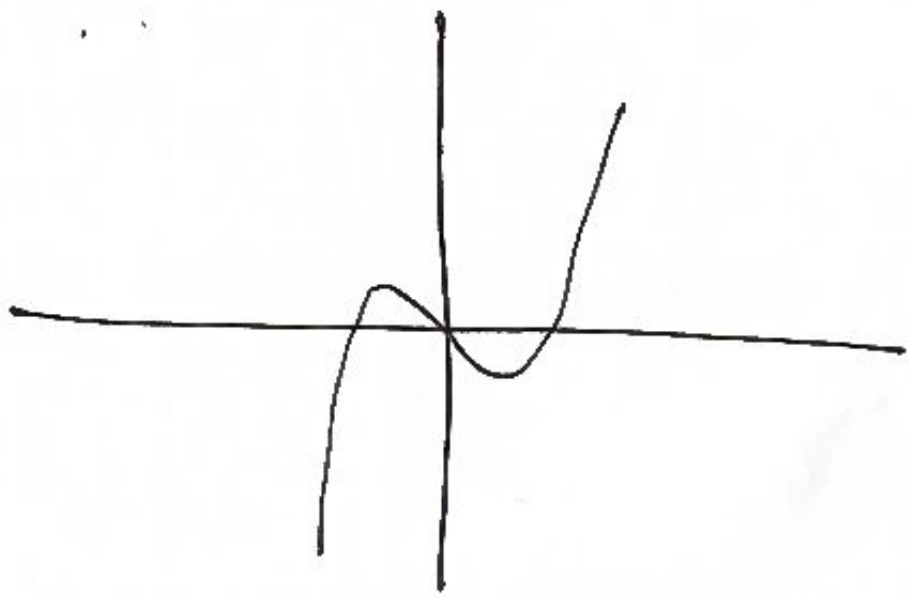


local minima

but doesn't have global

local and global maxima

$y = x^3 - 3x$



D- 08.07.2025

Economics

We should study economics for two reasons:

a. understand global phenomena
b. make wiser decision regarding financial and personal life

Def:

Economics is the study of turning limited (scarce) resources into valuable goods and services, and distributing them among people in the most efficient way.

~~Every~~ Economics ~~phenomena~~ revolves around the concepts of scarcity (limited resources) and efficiency (where we try to find the best combo of quality and quantity)

So, economics study ⟶ resources
production
distribution

Demand and supply are two core concepts.

Greatly disrupted by global or local events such as natural calamities, COVID-19, Russia-Ukraine War

- The supply chain took a heavy blow in COVID-19 time; so the supply ↓↓ and the prices of goods and services ↑↑

- The Russo-Ukraine war came the US, EU imposed sanctions on Russian oil, gas, money etc. And Ukraine got heavily hit. Resulted in inflation in the price of oil, gas, food and others goods, services related to them.

Economics

- Microeconomics
  - Focuses more on individuals
  - Discussion on monopoly and market competition
  - Our syllabus mostly focuses on microeconomics.
- Macroeconomics
  - Focuses on the larger pictures on a national or global level

Discussed some ideas on interest rate

- Interest rate ↑ savings encouragement
  discourages money flow
- Interest rate ↓ investment increases
  but savings ↓

This was the first class and more an appetizer/ introductory class rather than a serious one.

Linear Algebra

Introductory class. Started with some very elementary things. Topic was "System of Linear Equations."

$ax+by=c$ (a, b not both 0)

$ax+by+cz=d$ (a, b, c not all 0)

x, y, z ~~a,b,c~~ $\longrightarrow$ unknown / variables

$a, b, c \longrightarrow$ constants

$a_1x_1+a_2x_2+\cdots\cdots+a_nx_n=b$ —— (i)

$a_1x_1+a_2x_2+\cdots\cdots+a_nx_n=0$ —— (ii)

(ii) is a homogenous linear equation ; as R.H $=0$

Def:

A finite set of linear equations is called a system of linear equations or simply, linear system.

Eg:

$3x+y=3$ —— (i)

$2x-y=4$

$4x_1-x_2+3x_3=-1$

$3x_1+x_2+4x_3=-4$ —— (ii)

1.1 – 1.7 in the syllabus of chapter-01

① $\left.\begin{array}{l} x-y=1 \\ 2x+y=6 \end{array}\right\}$ $x = 7/3,\ y = 4/3$

this linear system has only one solution.

② $\left.\begin{array}{l} x+y=4 \\ 3x+3y=6 \end{array}\right\}$ has no solution

③ $\left.\begin{array}{l} 4x-2y=1 \\ 16x-8y=4 \end{array}\right\}$ has infinitely many solutions

$y = 2x - 1/2$

if $x = t$; $y = 2t - \frac{1}{2}$

here, t is an arbitary value; known as a parameter

A general linear system in their unknowns $x_1, x_2, \ldots, x_n$:

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$
$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$
$$\cdots\cdots\cdots$$
$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m$$

A solution of a linear system m in n unknowns $x_1, x_2, \ldots, x_n \longrightarrow S_1, S_2, \ldots, S_n$

$$x_1 = S_1,\ x_2 = S_2, \ldots, x_n = S_n$$

$S_n \longrightarrow$ order n duple $\quad n = 2$; S_2 = ordered pair

$n = 3$; S_3 = ordered triple

Probability

Introductory class. Many of the concepts are pending.

- Experiment → ~~to an exen~~ obtains measurement or observation
  → unit of an experiment is known as trial
- Outcome → result of an experiment
- Sample space → Set of all possible outcomes of an experiment; denoted by S or Ω

① ~~Two~~ One die is thrown:

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

② Two coins are thrown:

$$S = \{HH \quad HT \quad TH \quad TT\}$$

③ One die and a coin is thrown

$$\Omega = \{1H \quad 1T \quad 2H \quad 2T \quad 3H \quad 3T \quad 4H \quad 4T \quad 5H \quad 5T \quad 6H \quad 6T\}$$

However, some basic concepts of counting, permutation, combination were discussed.

Making a password with 2 letters, 1 digit ?

$26 \cdot 26 \cdot 10 = 6760$ combinations are possible

Some differences between permutation and combination

Permutation	Combinations
• Order matters	• Order does not matter
• $_nP_r = \frac{n!}{(n-r)!}$	• $_nC_r = \frac{n!}{r!(n-r)!}$
n objects taking r objects at a time	$\therefore {}_nC_r = \frac{_nP_r}{r!}$

so, here BC and CB is the same thing, so we may get fewer possible scenarios than permutation

Orders matter (Permutation):

- Arrangement: Lines, bookshelf books, password
- Position: Specific roles assignments or ranking the competitors.
- Sequence: 1st, 2nd, 3rd position in a race.

Order doesn't matter (Combination):

- ~~Arrangement~~
  - Selection / Group: Choosing random members for a committee, picking lottery numbers, selecting a hand of cards.
  - Collection: buying a set of fruits, choose toppings for a pizza.
  - Subset: Picking a group of items without regarding the sequence.

Q) In a race with 10 runners, how many ways one can recieve gold, silver, bronze medals?

→ Order matters, so permutation

$$P(10,3) = \frac{10!}{(10-3)!} = 720 \text{ ways}$$

Q) A club has 10 members. In how many ways, a 3 person committe can be formed?

→ Order doesn't matter, so combination

$$10C3 = C(10,3) = 120$$

$$= \frac{10!}{(10-3)!\,3!}$$

Q) How many different ways a 4-digit pin can be created?

(a) If repetition allowed $\longrightarrow$ $10 \times 10 \times 10 \times 10 = 10000$

(b) And if repetion is not allowed $\longrightarrow$

$10 \times 9 \times 8 \times 7 = 5040$

Q) You are making a cake. 3 ingredients, 3 icings, 15 toppings?

$\longrightarrow$ Counting

$\hookrightarrow$ May be possible in $3 \times 3 \times 15$ ways.

Q) In how many ways 5 books be arranged in a shelf?

$\longrightarrow$ $P(5,5) = 120$ ways

Combining things

Q) Out of 5 consonants and 3 vowels, how many words of 3 consonants and 2 vowels can be formed?

$\longrightarrow$ ~~$5P3 \times 3P2 = 360$~~ ~~$5C3 \times 3C2$~~ $5P3 \times 3P2 = 360$

~~Q) A chess club~~

Q) PHONE $\longrightarrow$ Organize the words so that vowels come together?

$\longrightarrow$ $4! \times 2! = 48$ ways

Q) PHONE → Organize the words so that consonents come together ?

→ $3! \times 3! = 36$

Permutation with repetation:

- Applicable only if all objects used

Formula: $\frac{{}^{n}P_{n}}{p!\,q!}$

Combination with repetation:

$$^{n}C_{r} = \frac{(n+r-1)!}{r!\,(n-1)!}$$

D – 09.07.2025

101

Statistics is a branch of science that deals with the collection, processing and interpretation of masses of facts & figure to predict unseen scenario.

Descriptive Statistics

Known as Inferential Statistics

Statistics

↓

Applications of it briefly discussed it applied statistics

↓

an inter-disciplinary subject

↓

Our core responsibility will be to help in other's works by our knowledge.

Statistics is about generating description to get a general idea about something.

Programming – C

- Linux commands
- A basic hello.c program execution

Differential Eq^n

- independent variable

  'dependant variable's dependency on an independent variable

Example of a differential equation:

$$\frac{d^2y}{dx^2} + xy\left(\frac{dy}{dx}\right)^2 = 0$$

Differential equation of $y = ax$

$\longrightarrow$ $y = ax \longrightarrow \frac{dy}{dx} = a$

$\therefore \quad y = \left(\frac{dy}{dx}\right)x$

Differential equation of $y = Ae^{x} + Be^{-x}$

$\longrightarrow$ $y = Ae^{x} + Be^{-x}$

$\Rightarrow \frac{dy}{dx} = Ae^{x} - Be^{-x}$

$\Rightarrow \frac{d^2y}{dx^2} = Ae^{x} + Be^{-x}$

$\Rightarrow \frac{d^2y}{dx^2} - y = 0$

Differential equations
- Ordinary (Course syllabus)
- Partial

Order and degree of a differential equation:

- $\frac{d^2y}{dx^2} + xy\left(\frac{dy}{dx}\right)^2 = 0$

  is an ordinary differential equation.

  Order $\longrightarrow 2$

  Degree $\longrightarrow 1$

Ordinary differential equation
- linear equation
- non-linear equation

- degree of the equation : 1
- no transcendental functions of y (the dependant variabl
  (trig, exponential or logarithmic)
- no products of y or its derivatives ; eg:
  - $\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)$
  - y^2, y^3 etc.
  - $y\left(\frac{dy}{dx}\right)$

If the criterias don't match, it's non-linear equation

(a) $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y^2 = 0 \longrightarrow$ non-linear (products of y)

(b) $\frac{d^2y}{dx^2} + 5y\frac{dy}{dx} + 6y = 0 \longrightarrow$ non-linear $\left(y \cdot \frac{dy}{dx}\right)$ ~~(products of~~

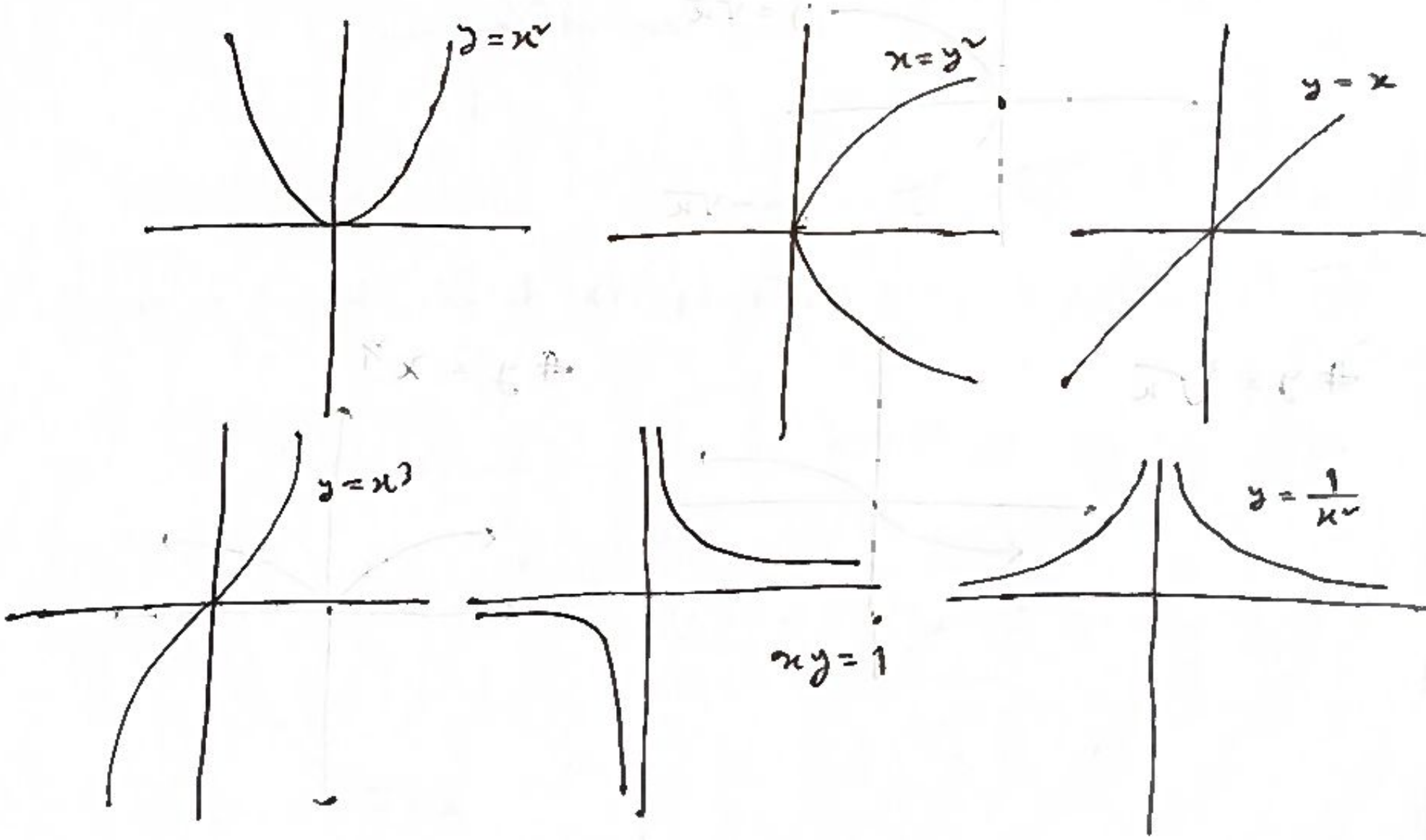
(c) $\frac{d^2y}{dx^2} + y\sin x = 0 \longrightarrow$ linear diff eq^n

(d) $\frac{d^4y}{dx^4} + x^2\frac{d^2y}{dx^2} + x^3\frac{dy}{dx} = xe^x \longrightarrow$ linear diff eq^n

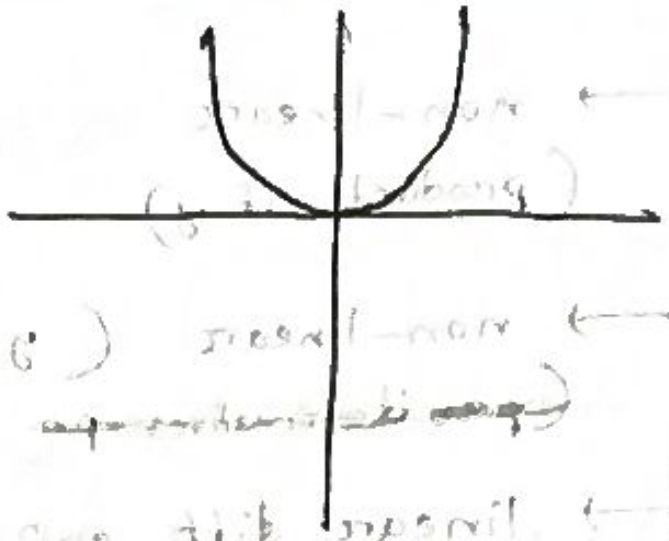
(e) $\frac{d^2y}{dx^2} + x\sin y = 0 \longrightarrow$ non-linear equation

Differential Calculus

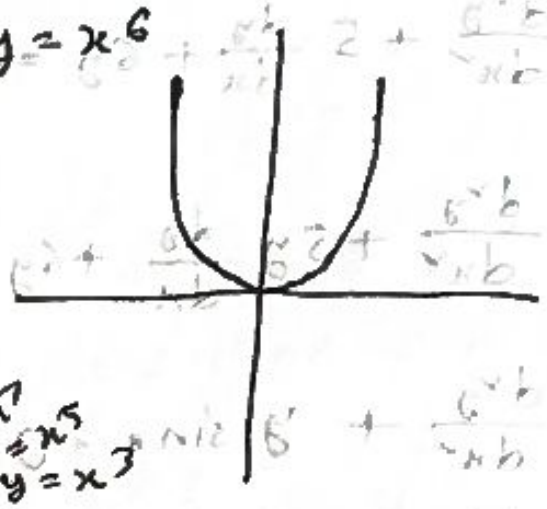
Common graphs:



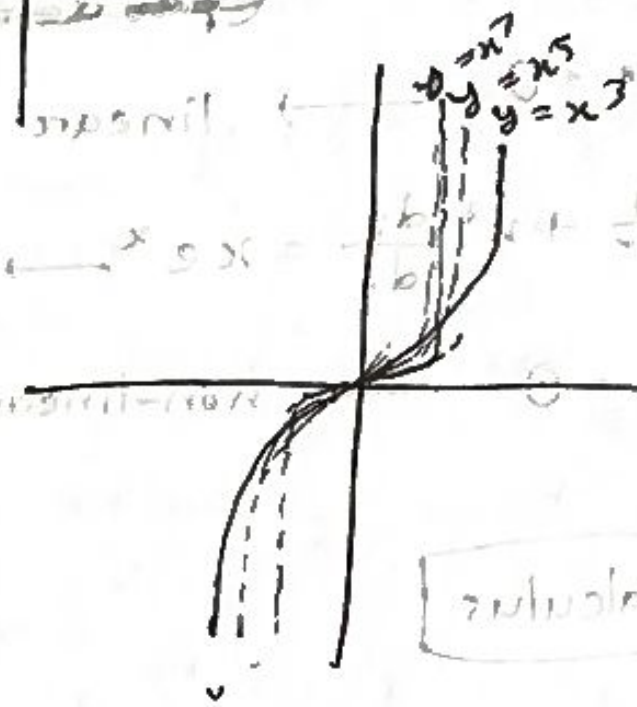
$y = x^4$



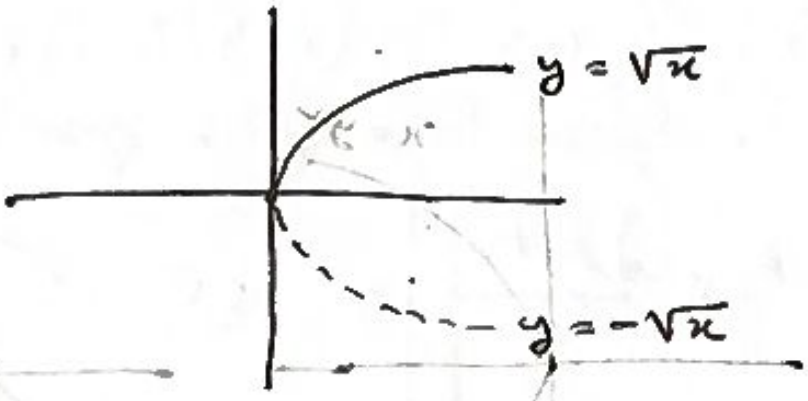
$y = x^6$



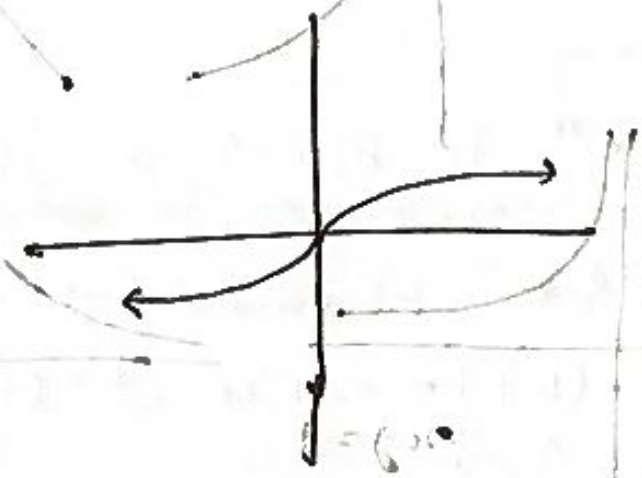
$y = x^5$, $y = x^7$, $y = x^3$



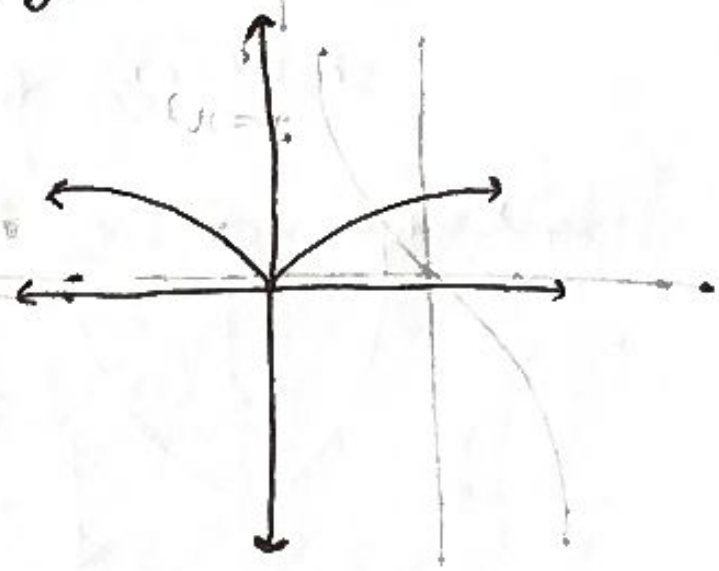
$y = \sqrt{x}$; $y = -\sqrt{x}$



$y = \sqrt[3]{x}$



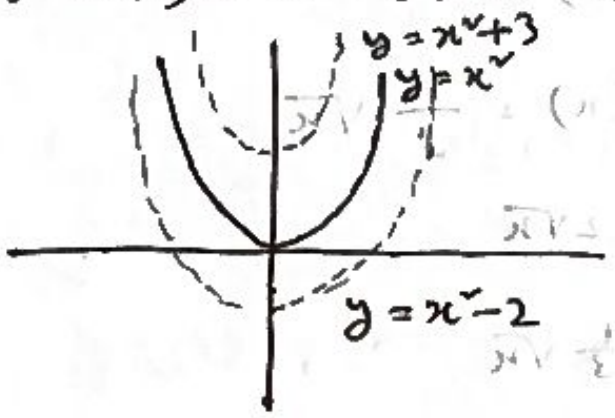
$y = x^{2/3}$



Building New Graphs:

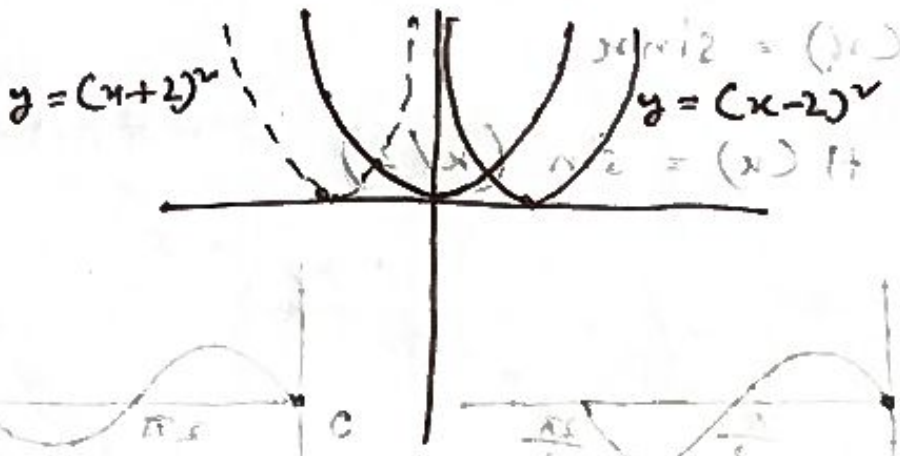
$y = F(x)$; $G(x) = F(x) + k$; (Vertical Shift)

(a) $y = x^2$, $G(x) = x^2 + 3$, $H(x) = x^2 - 2$



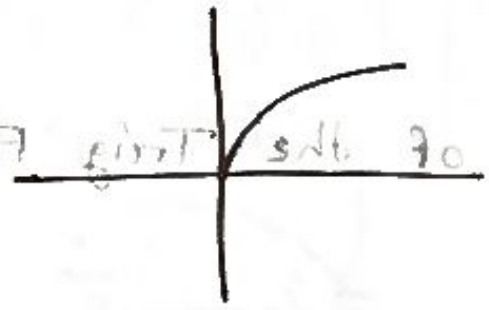
$y = F(x)$: $G(x) = F(x-h)$; (Horizontal Shift)

(a) $y = x^2$; $G(x) = (x-2)^2$; $H(x) = (x+2)^2$

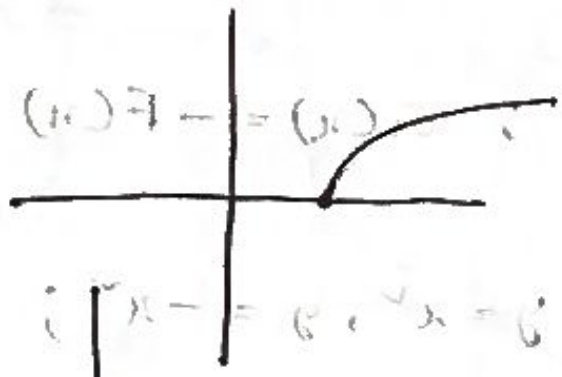


Q. Create a rough sketch of $y = \sqrt{x-2} - 3$

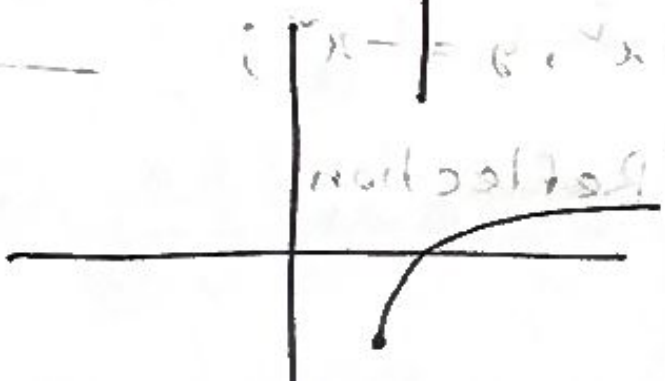
a. $y = \sqrt{x}$



b. $y = \sqrt{x-2}$



c. $y = \sqrt{x-2} - 3$

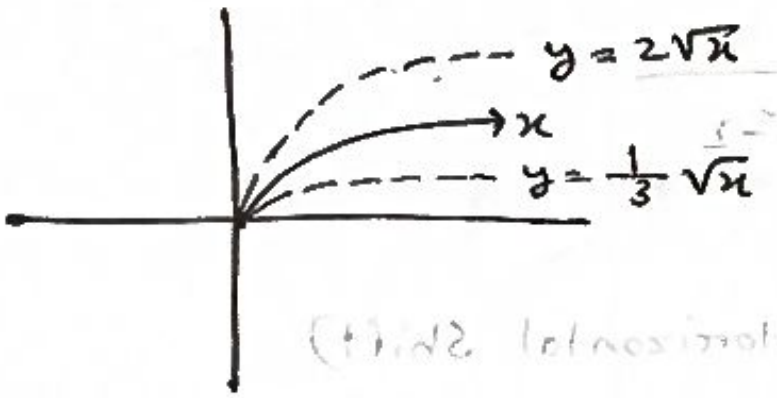


~~# y = H (kx)~~

~~y = sinx~~

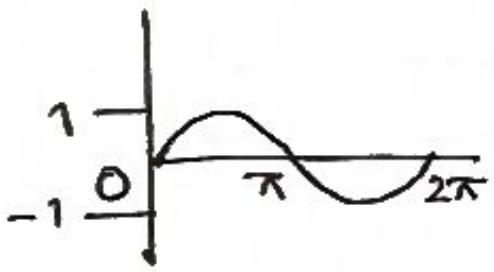
$y = f(x)$; $y = k\, f(x)$

$H(x) = 2\sqrt{x}$, $I(x) = \frac{1}{3}\sqrt{x}$

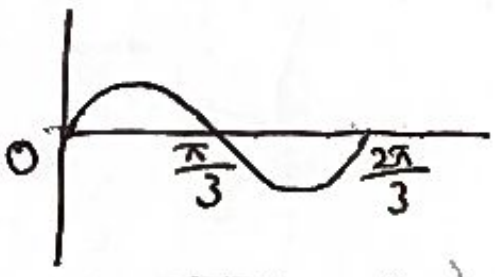


$y = F(kx)$; $F(x) = \sin x$

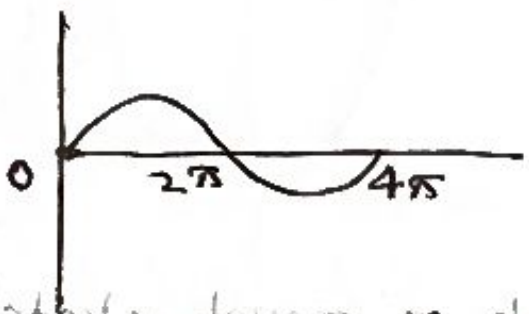
$G(x) = \sin(3x)$, $H(x) = \sin(x/2)$



$F(x) = \sin x$



$G(x) = \sin(3x)$



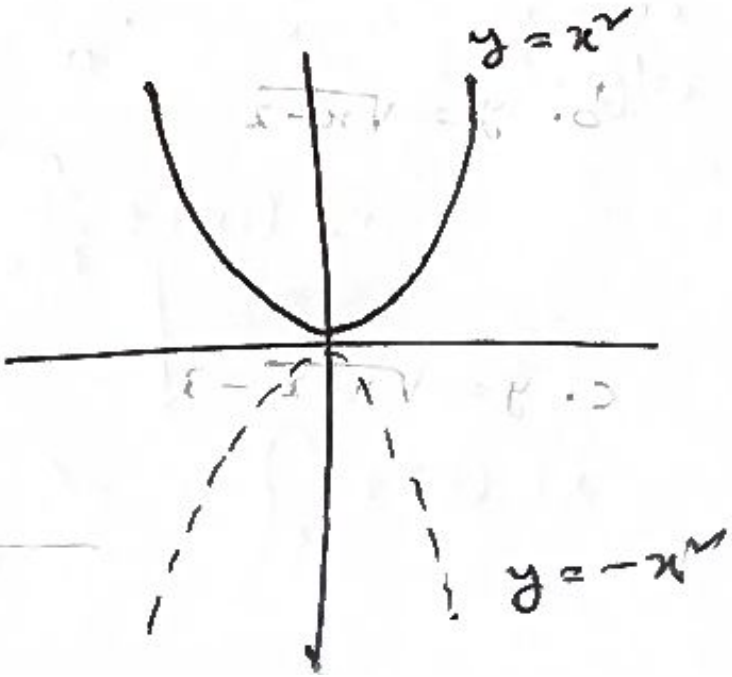
$H(x) = \sin(x/2)$

* Amplitude & Period of the Trig Functions

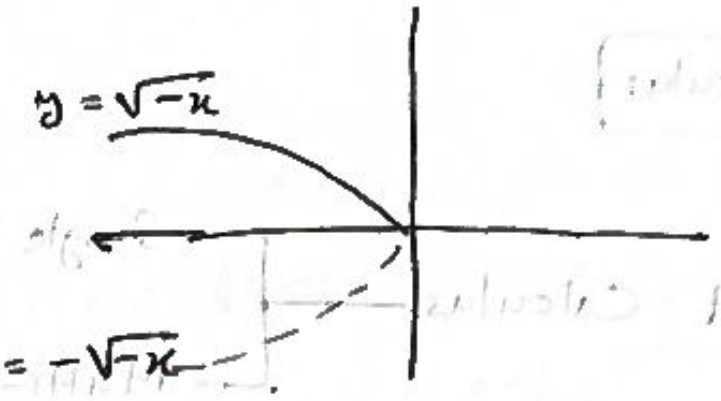
$y = F(x)$; $G(x) = -F(x)$

(a) $y = x^2$, $y = -x^2$;

Reflection



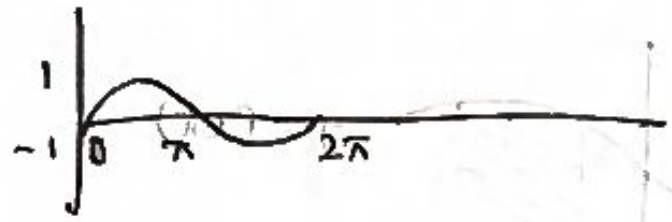
(b) $y = \sqrt{-x}$, $y = -\sqrt{-x}$



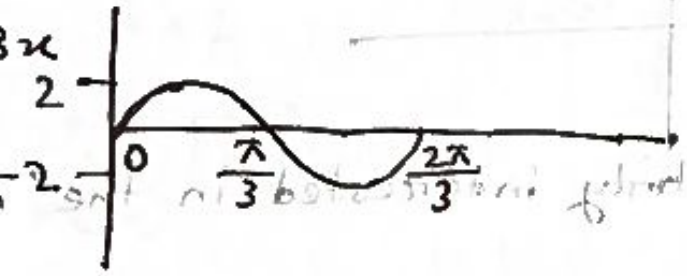
Q) Draw a rough sketch of $2 \sin 3\left(x - \frac{\pi}{4}\right)$

→

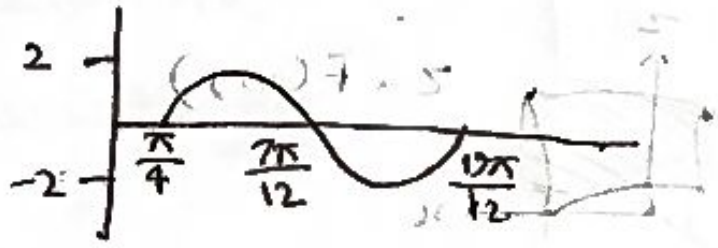
(a) $\sin x$



(b) $2 \sin 3x$



(c) $2 \sin 3\left(x - \frac{\pi}{4}\right)$



Q) Draw a rough sketch of $y = \sqrt{1-x} + 2$

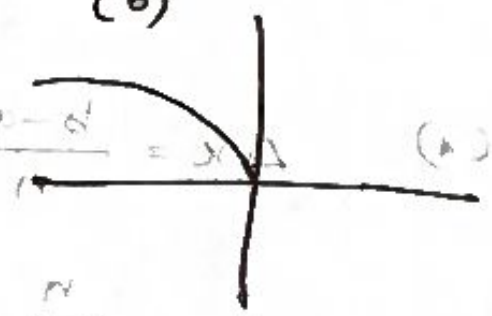
→ (a) $y = \sqrt{-(x-1)} + 2$

(b) $y = \sqrt{-x}$

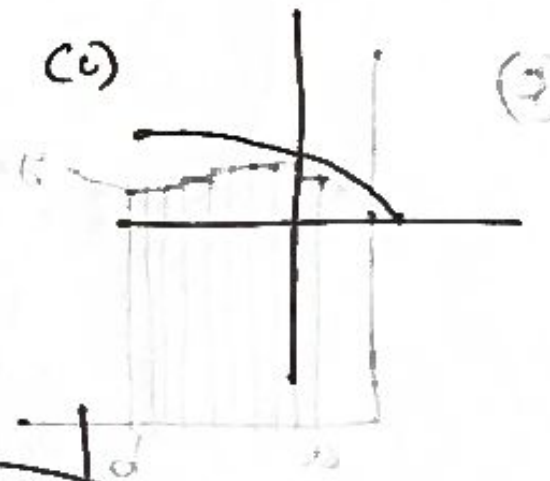
(c) $y = \sqrt{-(x-1)}$

(d) $y = \sqrt{-(x-1)} + 2$

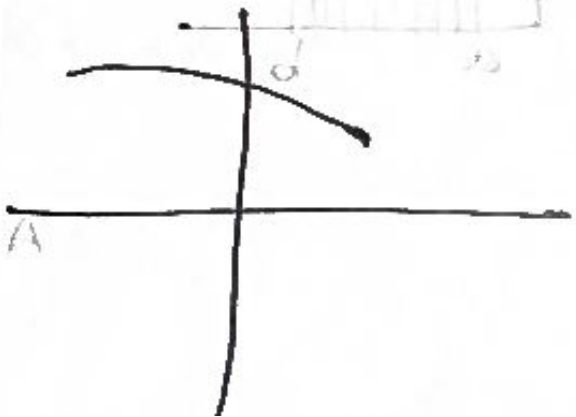
(b)



(c)



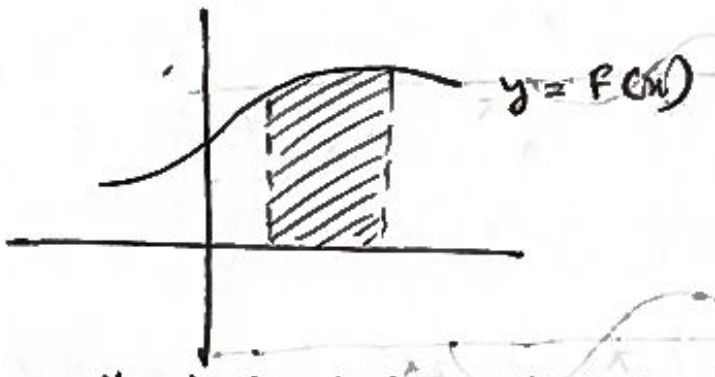
(d)



Integral Calculus

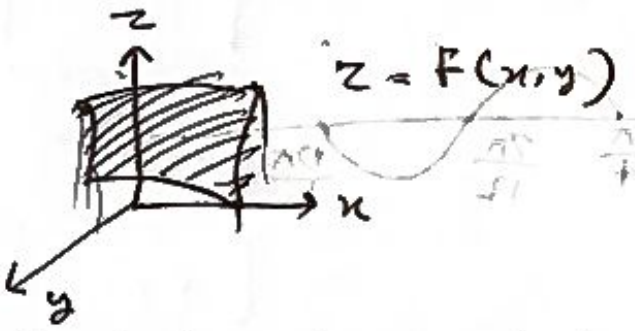
Integral Calculus
- Single Variable
- Multi-variable

(a) Graph of a single-variable function $\longrightarrow$ 'Curve'



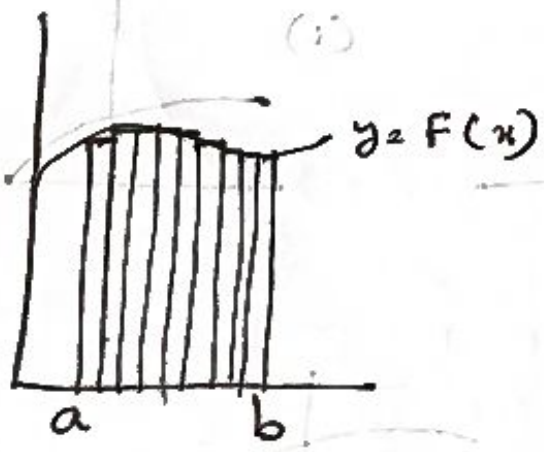
We are particularly interested in the 'Area' here.

(b) Graph of a multi-variable function $\longrightarrow$ Surface



We are particularly interested in the 'Volume' here.

(c)



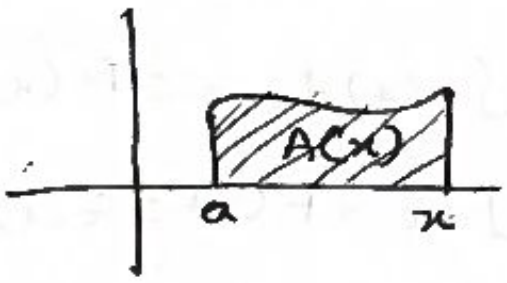
$$\Delta x = \frac{b-a}{n}$$; n = number of rectangles

$$A_n = \sum_{i=1}^{n} f(x_i)\,\Delta x_i$$

$$\text{Area} = \lim_{\substack{n \to \infty \\ \text{or } \Delta x \to 0}} A_n = \int_a^b f(x)\,dx$$

Anti-derivative:

Let's, $y = f(x) \geq 0$ is a continuous function on $[a, x]$

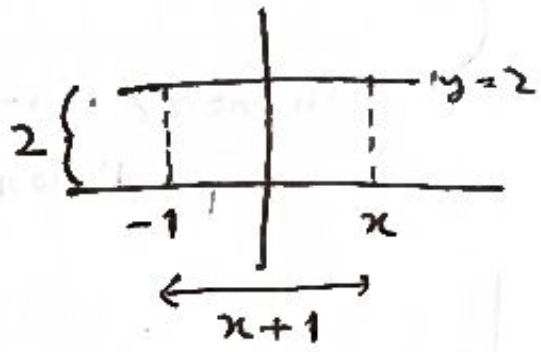


- $A(a)$ are area of line a is zero (0)

And $\boxed{A'(x) = F(x)}$

(a) $F(x) = 2$; $[-1, x]$

$\Rightarrow y = 2$



$A(x) = 2(x+1) = 2x + 2$

$A'(x) = \frac{d}{dx}(2x+2) = 2$

$= F(x)$

~~(b) $f(x) = x^2 = 4$; $[a, x]$~~

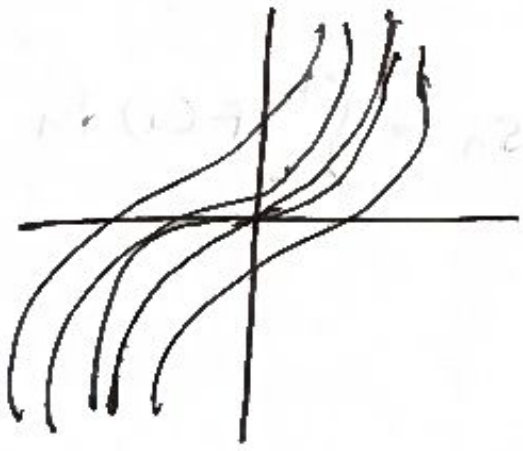
~~Area(x) = $\int_a^x$~~

D – 10.07.2025

$F(x)$ is an anti-derivative of $f(x)$ if $F'(x) = f(x)$

Integral curve:

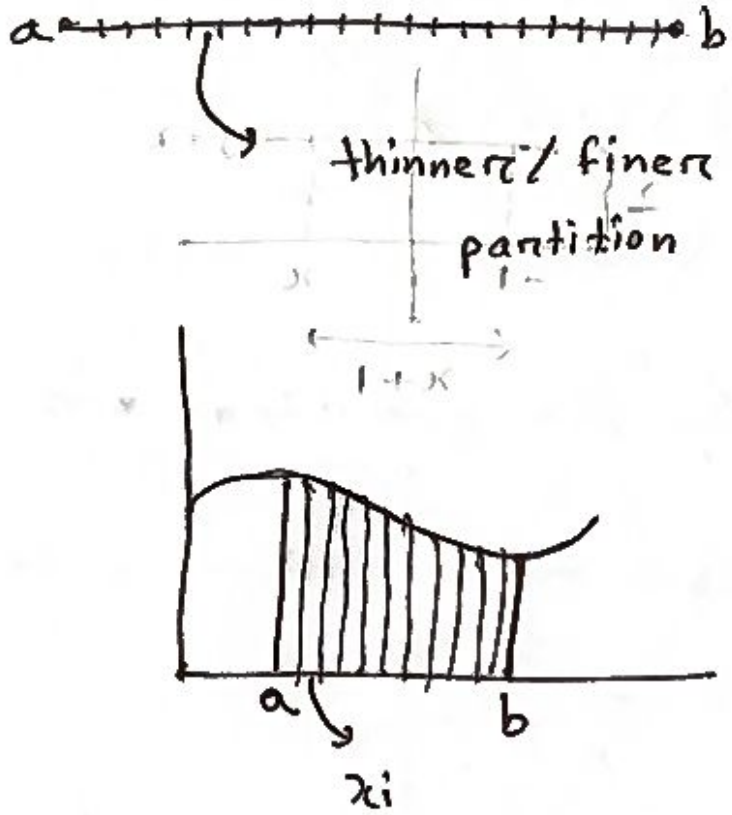
$$\int x^2 \, dx = x^3/3 + c$$



$\int c f(x)\,dx = c\int f(x)\,dx = c\,F(x) + k$

$\int [f(x) \pm g(x)]\,dx = F(x) \pm G(x) + k$

Refinement:



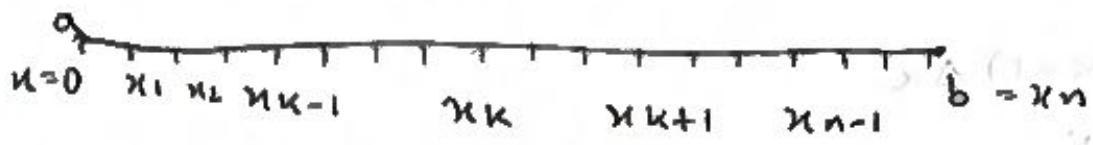
$x_i^* \in [x_{i-1}, x_i]$

$S_n = \sum_{i=1}^{n} f(x_i^*)\,\Delta x \quad ; \quad \Delta x = \frac{b-a}{n}$

$n \to \infty \; ; \; \Delta x \to 0$

Riemann sum

$\lim_{\substack{n\to\infty \text{ or} \\ \Delta x \to 0}} S_n = \int_a^b f(x)\,dx$



$[x_{k-1}, x_k]$

left point $= x_k^* = x_{k-1} = a + (k-1)\Delta x$

right point $= x_k^* = x_k = a + k\Delta x$

Midpoint $= x_k^* = \dfrac{x_k + x_{k-1}}{2}$

$= a + \left(k - \frac{1}{2}\right)\Delta x$

$1 + 2 + 3 + \cdots + n = \dfrac{n(n+1)}{2}$

$1^2 + 2^2 + 3^2 + \cdots + n^2 = \dfrac{1}{6} n(n+1)(2n+1)$

$1^3 + 2^3 + 3^3 + \cdots + n^3 = \left[\dfrac{n(n+1)}{2}\right]^2$

Q) Find the area between the graph of $f(x) = x^2$ and the interval $(0, 1)$

$\longrightarrow$ $a = 1,\ b = 0$ $\qquad \Delta x = \dfrac{1-0}{n} = 1/n$

Right point, $x_k^* = a + k\Delta x = 0 + k\Delta x = k/n$

$f(x_k^*)\Delta x = \left(\frac{k}{n}\right)^2 \times \frac{1}{n} = \dfrac{k^2}{n^3}$

$\therefore$ ~~Area~~ $S_n = \displaystyle\sum_{k=1}^{n} \left(\frac{k}{n}\right)^2 \times \frac{1}{n} = \frac{1}{n^3} \sum_{k=1}^{n} k^2$

$= \dfrac{1}{n^3} \cdot \dfrac{1}{6} n(n+1)(2n+1)$

$\therefore$ Area $= \displaystyle\lim_{n\to\infty} S_n = \lim_{n\to\infty} \frac{1}{6} \cdot \frac{1}{n^3} \cdot n^3 \left(1 + \frac{1}{n}\right)\left(2 + \frac{1}{n}\right)$

$= \dfrac{1}{6} \times 2 = \dfrac{1}{3}$ sq. unit

For left point : $x_k = a + (k-1)\Delta x$

$$= \frac{k-1}{n}$$

$$\therefore \quad f(x_k)\,\Delta x = \left(\frac{k-1}{n}\right)^2 \times \frac{1}{n} = \frac{(k-1)^2}{n^3}$$

$$\therefore \quad S_n = \sum_{k=1}^{n} \frac{(k-1)^2}{n^3}$$

$$= \frac{1}{n^3} \sum_{k=1}^{n} (k-1)^2$$

$$= \frac{1}{n^3} \cdot (n-1)(n-1+1)(2n-2+1)$$

$$= \frac{1}{n^3} \frac{1}{6} n (n-1)(2n-1)$$

$$\therefore \text{Area} = \lim_{n\to\infty} S_n = \lim_{n\to\infty} \frac{1}{6} \frac{1}{\cancel{n^3}} \cdot \cancel{n^3} \left(1-\frac{1}{n}\right)\left(2-\frac{1}{n}\right)$$

$$= \frac{1}{6} \times 2 = \frac{1}{3} \text{ sq. unit}$$